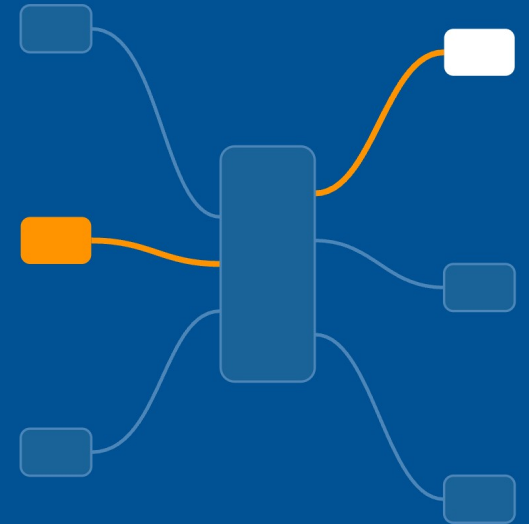




Essentials 3.0 & Dev Tools 1.6

What's new in the framework and the tools around it



AT A GLANCE

Two release candidates

PEPPERDASH ESSENTIALS

v3.0.0-rc.6

Compared with v2.42.4, the current release on main

180

commits

712

files changed

99

new unit tests

ESSENTIALS DEV TOOLS

v1.6.0-rc.1

Compared with v1.5.0, the current release

41

commits

135

files changed

6

headline features

Both are release candidates today. Everything that follows is a preview of what ships next.

PART 01

Essentials 3.0

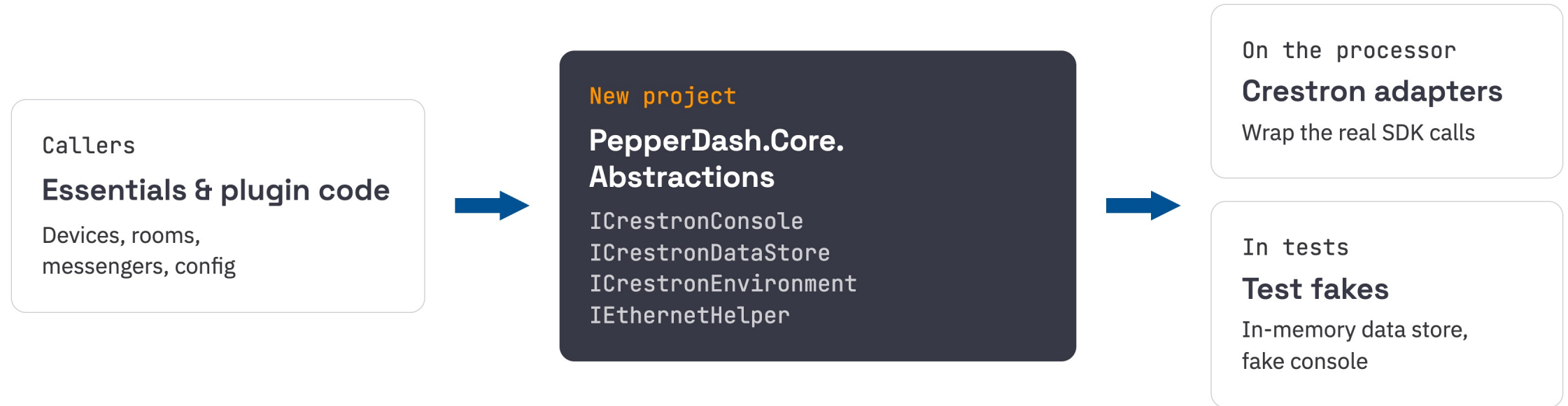
Platform · plugins · routing · Mobile Control · device capabilities ·
secrets · debugging



Now on .NET 8

Area	v2.x	v3.0
Runtime	.NET Framework 4.7.2	.NET 8 (C# 12)
Timers	Crestron CTimer	System.Timers.Timer
Threading & locks	CCriticalSection, CrestronInvoke	lock, Task
HTTP	Crestron HTTP clients	Built-in HttpClient
Automated tests	None	99 unit tests, 3 projects
Legacy APIs	30+ [Obsolete] members	Removed

Testable without a processor



99 unit tests across three new xUnit projects, run in CI on every build

Safer plugin loading



.NET 8 compatibility check

Plugins can declare [Net8Compatible]. Incompatible plugins, and anything that depends on them, are reported with the reason.



Version check against config

At startup, loaded Essentials and plugin versions are compared with the config's versions node. Mismatches are logged as warnings.



Startup errors you can find

Initialization exceptions are captured and available from the console and the web API.

[getinitializationexceptions](#)

Failed factories now log as errors, not "unknown type".

Under the hood: a rewritten plugin loader and IDeviceFactory model, designed to stay compatible with existing EssentialsPluginDeviceFactory<T> plugins.

Routing, consolidated

v2.x • 18 interfaces

`IRouting` `IRoutingNumeric`
`IRoutingNumericWithFeedback` `IRoutingWithFeedback`
`IRoutingWithClear` `IRoutingFeedback` `IMatrixRouting`
`IRmcRouting` `IRmcRoutingWithFeedback` `ITxRouting`
`ITxRoutingWithFeedback` `IRoutingSink`
`IRoutingSinkWithSwitching` `IRoutingInputsOutputs`
`IRoutingSlot` `IRoutingInputSlot` `IRoutingOutputSlot`
`IHasCurrentSourceInfoChange`



v3.0 • feedback first

IRoutingSinkWithFeedback

Switch, current input, input changed

IRoutingMidpoint

Inputs plus outputs

IRoutingMidpointWithFeedback

Current routes, route changed, clear route

+ **ICurrentSources**: the actual source device for each signal type

Routing you can watch and drive

Live feedback websocket

Streams a snapshot, then route, sink-input and layout changes as they happen.

`routingFeedbackSession`

Routing command API

Route a sink, switch a midpoint, or clear either. Validated, with a dry-run option.

`POST routingCommand`

Multiview layouts

Tile sinks, layout state and dynamic layouts are part of the routing graph.

`IRoutingSinkWithLayoutState`

Named routing slots

Matrix slots report their TX/RX device, online state and video sync.

`IHasNamedRoutingSlots`

Messengers built from interfaces

MessengerFactoryRegistry looks at the interfaces a device implements and attaches the matching messengers automatically.

29

new focused messengers, replacing the single VideoCodecBaseMessenger

Subscriptions removed; full-status requests can target specific devices.

CODEC

Call controls

Call history

Cameras

Camera mute

Camera presets

Layouts

Room presets

Self-view

Content sharing

Far-end content

Meeting info

Start meeting

Directory

Dialer

Codec info

PARTICIPANTS

Audio mute

Video mute

AUDIO CODEC

Codec info

Phonebook

Call status

DEVICE STATE

Standby

Ready

Warming / cooling

Privacy

Password prompt

Room schedule

ROUTING & ACTIONS

Midpoint feedback

Named routing slots

Device actions

Rooms, panels and deployment

Room combiner progress

Reports in-progress combinations and affected rooms, driving panel lockouts.

Per-panel lockout text

Set lockout messages for each touchpanel from system config.

`lockoutMessagesByTouchpanel`

User app deployment

mcUserApp zip files are deployed and extracted automatically.

Version awareness

Reports UI app and wrapper versions; the app version is read from disk.

Local UI development

Point panels at a local dev server while building React UIs.

`DevelopmentServerAddress`

Light & dark logos

Custom logo URLs for light and dark backgrounds.

New device capabilities

Room plugins control devices through the framework, instead of referencing each device plugin directly.

Recording

`IHasRecordingControl`
`IHasRecordingPause`
`IHasRecordingInfo`
`IHasRecordingSchedule`
`ICourtSession`

Audio & DSP

`IDtmfDecode`
`IDspPresetsDirectRecall`
`IAudioCodecPhonebook`
`IDialerCallStatus`

Codec & display

`ICodecCallControls`
`IDisplayCurrentInput`
`IAnnotationStream`
`IHasPolling`

Privacy & sharing

`IHasCameraPrivacy`
`IHasWirelessSharingControl`

Per-person setups

`IHasUserPresets`

Room settings per person, from a profile store

Also new

Richer `IHasWebView` events ·
CustomProperties on metadata items ·
`IsCohost` and `VideoMuteFb` on participants

Secrets manager



Write-only by design

Values can be set, replaced or deleted, but never read back, and they're kept out of the logs.



Bulk-friendly

Apply many secrets from one file, or download a template to start from.



Knows what it manages

A separate index tracks which data store keys were written through the API.

NEW WEB API ROUTES

secrets

List managed and other keys

secrets/providers

Available secret providers

secrets/command

Set, update or delete one secret

secrets/bulk

Preview or apply many at once

secrets/template

A starter file

Built for debugging



Dev Tools ships inside Essentials

Bundled, unpacked at startup and served straight from the processor.



Debug sessions find the right network

Session URLs list every reachable network, LAN or control subnet.
Fixes the "Invalid Value" host.



Config schemas

JSON schemas for Essentials and Mobile Control configs, for validation and autocomplete.

NEW CONSOLE COMMANDS

> debugappurl

Prints the Dev Tools URL

> getinitializationexceptions

Lists errors from startup

NEW WEB API ROUTES

routingFeedbackSession

routingCommand

secrets/*

initialization exceptions

PART 02

Dev Tools 1.6

Live routing · route control · secrets · built-in help

The routing page comes alive

Live routing feedback

Needs v3

The diagram updates as routes change. A Live/Offline badge, one-click reconnect, and live-only routes drawn dashed.

Signal path tracing

Click a line to light up its full path. Click a device to see every path through it. Switchers show their active connections.

Multiview layout panel

A floating panel shows each tile's live source. Click a tile to trace its path.

Route from the diagram

Needs v3

Click a port, pick a signal type and a source, or clear it. Ports show pending until the processor confirms.

Secrets and built-in help

Secrets page

- Add, replace or delete; values are never shown
- Download a template, or bulk-import a file with a preview first
- Other data store records listed separately, with overwrite confirmation
- Appears only when the processor offers the secrets API

Built-in help

- Help link in the top nav and on the login screen
- Docs shown in the app: Tutorials, How-to, Reference, Explanation
- New guides: changing routes, tracing signal paths, managing secrets

Fixes and polish

Fixes

- No more duplicate debug console messages
- Console tries the server's main URL first, then the fallback
- Switching apps cleanly ends the old debug session
- Bad websocket URLs fall back or report an error
- Routing diagram no longer jumps on live updates

Improvements

- Show/hide password on login
- Clear help when a certificate blocks live feedback
- Keyboard navigation for tiles, panels and Secrets
- "Connecting..." state on Start Debug Session
- Versions table scrolls with sticky headers

TOOLING

ESLint 10 + Prettier 3

Much larger test suite

Requires Node 20.19+

WHAT'S NEXT

Road to release

PepperDash®

Now

Release candidates

Essentials v3.0.0-rc.6
Dev Tools v1.6.0-rc.1

September - October 2026

Testing on real systems

Live at 2 client's sites already

Sometime in October?

3.0.0 and 1.6.0 final

General availability



Try the RCs on a test system, starting with the demo

[PepperDash/EssentialsDemoConfig](#)



Plugin authors: load your plugin on v3 and check for compatibility warnings



Report issues on GitHub

[PepperDash/Essentials](#)
[PepperDash/essentials-devtools](#)